

EXAM 7 – SPRING 2015

13. (1.75 points)

Given the following information for a reinsurer as of December 31, 2014 (\$000):

Accident <u>Year</u>	<u>Earned Premium</u>	<u>Earned Risk Pure Premium</u>	<u>Adjusted Premium</u>	<u>Aggregate Reported Loss</u>	<u>Aggregate Loss Report Lag</u>
2011	10,000	10,000	9,000	6,000	90%
2012	11,500	11,000	12,000	5,000	70%
2013	12,500	12,000	11,000	2,000	40%
2014	14,000	13,000	13,000	4,000	30%

a. (1.25 point)

Using the Stanard-Bühlmann Method, estimate the IBNR for all accident years combined.

b. (0.5 point)

Identify and briefly explain the need for the modification made to earned risk pure premium to derive adjusted premium.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 13					
TOTAL POINT VALUE: 1.75				LEARNING OBJECTIVE: A13: Calculate ceded loss reserves using appropriate methods.	
SAMPLE ANSWERS					
Part a: 1.25 points					
Sample Answer 1					
	Adjusted Premium	Aggregate Loss Report Lag		Total Reported Loss	
	9,000	90% =9,000*90%		6,000	
	12,000	70%		5,000	
	11,000	40%		2,000	
	13,000	30%		4,000	
	Used Up Premium	24,800		17,000	=6k+5k+2k+4k
	ELR	=17,000/24,800	=68.55%		
	Expected Loss	= ELR * Adj Prem * (1-Lag)			
	2011	0.61695			
	2012	2.4678			
	2013	4.5243			
	2014	6.23805			
	13.847	Million			
Sample Answer 2					
	Adjusted Premium	Aggregate Loss Report Lag		Total Reported Loss	
2011	9,000	90% =9,000*90%		6,000	
2012	12,000	70%		5,000	
2013	11,000	40%		2,000	
2014	13,000	30%		4,000	
	Used Up Premium	24,800		17,000	=6k+5k+2k+4k
	ELR	=17,000/24,800	=68.5%		
	Expected Loss	= ELR * Adj Prem * (1-Lag)			
	2011	0.6165			
	2012	2.466			
	2013	4.521			
	2014	6.2335			

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13.837 Million
Part b: 0.5 point
<i>Sample Answer 1</i> Adjust the ERPP to on-level premiums so that the loss ratio for each AY would be comparable. <i>Sample Answer 2</i> You need to adjust for varying rate levels because the SB method assumes a constant ELR across all years. <i>Sample Answer 3</i> Current rate level. We want all of the premiums to be on the same level because we are calculating a single ELR from all years. <i>Sample Answer 4</i> Adjusting for different rate levels is necessary as the losses that have yet to emerge will do so at today's cost levels. The adjusted premium should therefore give us a better estimate of the ELR in order to calculate IBNR.
EXAMINERS' REPORT
Candidates were required to complete an IBNR estimate using the Stanard-Bühlmann technique. The vast majority of candidates were able to do this successfully.
Part a
Candidates generally performed well on this part. Common mistakes included using Earned Risk Pure Premium instead of Adjusted Premium, as well as calculating IBNR by subtracting the reported losses from the expected ultimate losses.
Part b
The bulk of candidates correctly identified Current Rate Level as the required adjustment. Discussion of trend and other similar terms/concepts also earned credit. Candidates who mentioned adjusting for expenses and other similar terms/concepts did not receive any credit, as these adjustments are already included in the Earned Risk Pure Premium.